

Pennsylvania Health Insurance Audit — Carrier Filing Pipeline

Data-engineering foundation for a larger public information campaign on Pennsylvania's 2026 individual-market rate increases and the expiration of Enhanced Premium Tax Credits (EPTCs). This repo currently covers **carrier rate filing validation, ingestion, and aggregation** — the earliest stage of a much larger proposed architecture (see "Build-Out Roadmap" below).

What's actually built and tested right now

Piece	File	Status
PA county/FIPS/rating-area crosswalk	<code>bootstrap/pa_crosswalk_data.py</code>	Single source of truth. Self-validating on import (uniqueness, range, count). Verified against CMS's official PA geographic rating area listing and real PID filing PDFs.
Crosswalk generators	<code>bootstrap/02_crosswalk.py</code> , <code>scripts/bootstrap_data.py</code>	Both derive from the SSOT above — no duplicated county data.
Carrier filing structural audit	<code>bootstrap/04_audit.py</code>	Fast pass/fail gate: schema, rate bounds, rating-area validity. No hardcoded carrier roster (PID's real filer list changes every cycle).
Carrier + termination ingestion	<code>bootstrap/05_ingest_carrier_extensions.py</code>	Weighted rate-change aggregation by area/carrier; termination-rate aggregation by area. Handles <code>"1-9"</code> ranges and <code>"6;7;9"</code> lists in <code>primary_rating_areas</code> .
Orchestrator	<code>bootstrap/00_run_all.sh</code>	Runs <code>scripts/bootstrap_data.py</code> then <code>bootstrap/0N_*.py</code> in order, fails fast.
Tests	<code>tests/*.py</code>	39 passing, including a regression test against the exact file <code>bootstrap_data.py</code> generates.

Run it: `./bootstrap/00_run_all.sh` from the repo root.

Known gap: no real county-termination dataset exists yet (`05_ingest` will fail at that step until one is sourced) — tracked, not a bug.

Data provenance — verified vs. proposed

A pattern from this project's build process worth stating plainly: multiple pasted drafts of *code* (a carrier auditor, a crosswalk table) turned out to contain fabricated or corrupted data that looked plausible at a glance. The same scrutiny needs to apply to *content* data going forward, so this section separates what's actually been checked from what hasn't:

- **Verified against independent reporting** (news coverage, CMS/PID filings, cross-checked during this build): peak 2025 Pennie enrollment (497,000), OE 2026 close (486,000), March 24 2026 enrollment (471,442, exact match), ~102% average premium increase, hardest-hit cohorts (ages 55-64, 150-200% FPL), 2026 statewide weighted-average rate increase (21.5%).
- **The campaign's own policy analysis, not an external fact to verify:** the \$50M investment ask, the 44,000-enrollee restoration projection, the 3:1-4.6:1 ROI range, the 9-12% broad premium reduction estimate. These are proposals this project is making, not numbers to fact-check against a source — document the derivation (e.g. a future `docs/methodology/roi-calculation.md`), don't cite them as independently confirmed.
- **Not yet checked:** county-level termination rates (e.g. a specific county leading at 24%), the 12-item consumer-info-gap list, the 424-respondent PHAN survey figures, specific congressional-district breakdowns. These may well be accurate — they just haven't been independently spot-checked in this repo's history the way the enrollment and premium figures have. Check primary sources before they ship in a press release or legislative brief.

Build-Out Roadmap

The full target architecture is a public-facing campaign repo — API, interactive charts, PDF policy briefs, press kit, outreach materials. That's a large undertaking; building it in stages, each one testable before the next starts, rather than scaffolding everything at once:

Stage 1 — Data engineering foundation (current)

Crosswalk SSOT, carrier filing audit, carrier + termination ingestion.  built, tested.

Stage 2 — Analysis layer (`analysis/` , `data/processed/` , `src/analysis/`)

Consumer info-gap analysis, enrollment trend modeling, demographic segmentation, ROI derivation — built on Stage 1's validated data plus new sourced datasets (enrollment timeline, survey results, county terminations). Each new dataset gets the same provenance treatment as the section above before it's treated as ground truth downstream.

Stage 3 — Visualization & API (`assets/visualizations/` , `src/api/`)

Vega-Lite chart specs, static exports, a FastAPI service exposing audit metrics. Depends on Stage 2's processed data existing and being validated.

Stage 4 – Deliverables (`deliverables/audit-report/` , `policy-briefs/` , `outreach-materials/`)

Markdown source documents compiled into the full audit report, fiscal note, legislative brief, consumer white paper. Depends on Stage 2/3 for figures and charts – don't hand-write numbers into these that should be pulled from validated data.

Stage 5 – Publication infrastructure (`.github/workflows/` , `Dockerfile*` , `press-kit/`)

CI, containerized API deployment, PDF generation pipeline, press materials. Last, because it automates a pipeline that needs to exist first.

A full directory blueprint and file schemas for later stages already exist as a planning document – not built yet. Build the next stage when Stage 1's real-data gap (termination dataset) is closed, since Stage 2's demographic/geographic analysis depends on it.

Repo structure (current, not the full target)

```
pa-audit/
├── bootstrap/           # numbered pipeline stages + shared crosswalk SSOT
├── scripts/             # bootstrap_data.py – real data seed
├── tests/               # 39 tests, plus fixtures/ for retired synthetic data
├── data/
│   ├── public/         # crosswalk, carrier rates, rating-area reference
│   ├── private/        # generated reports, gitignored
│   └── README.md       # per-file data provenance detail
```